

What is
SharePoint?



SHAREPOINT E3

Foundation Guide

ABSTRACT

This guide documents a hands-on Microsoft 365 E3 SharePoint environment built to simulate workflows and develop foundational enterprise administration skills.

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Trula's Tech Trails Learning Lab

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EXECUTIVE SUMMARY

Purpose

The Tech Learning Lab environment has been established to develop foundational competencies in Microsoft 365 enterprise administration, with specific emphasis on Azure Active Directory (Azure AD) and SharePoint Online. This hands-on learning environment simulates real-world enterprise configurations to build practical skills prior to beginning professional responsibilities.

Key Learning Objectives

1. Azure Active Directory (Azure AD) Administration

- Understand that Azure AD is a cloud-based identity and access management service, distinct from traditional on-premises Active Directory Domain Controllers
- Note: Azure AD does not replace Windows Server Active Directory; organizations requiring domain controller functionality must deploy Windows Server environments
- Master Azure AD's core functions: user identity management, authentication, authorization, and security controls
- Implement security permission frameworks aligned with enterprise best practices

2. SharePoint Online Navigation and Administration

- Achieve confident beginner-level proficiency in SharePoint site navigation and administration
- Understand SharePoint's permission model and its independence from NTFS file system permissions
- Apply security best practices for document management and collaboration

Lab Implementation Plan

The learning environment will consist of eight user accounts, including one designated administrator account. This configuration will enable hands-on practice with:

- User provisioning and license assignment
- Permission inheritance and unique permission application
- SharePoint site creation and configuration
- Document library and folder structure implementation
- Security testing scenarios with varied permission levels

INITIAL SETUP OBSERVATIONS

User Provisioning and Configuration Discovery

During the initial configuration of the Tech Learning Lab environment, several important distinctions between on-premises Active Directory and cloud-based Azure Active Directory became apparent. These observations emerged organically during the user account creation process and highlight critical differences in administrative control, security enforcement, and policy management between the two platforms.

Primary Discovery: Password Policy Enforcement

While provisioning test user accounts (TestLab1, TestLab2, etc.), an attempt to implement simplified passwords for laboratory purposes revealed fundamental architectural differences between Windows Server Active Directory and Microsoft 365's Azure Active Directory. In previous on-premises Active Directory lab exercises, administrators possessed unrestricted ability to configure password policies, including the ability to reduce minimum password length to two characters for educational purposes. However, Azure Active Directory enforces a non-negotiable 8-character minimum password requirement that cannot be overridden regardless of administrative privileges.

This discovery prompted deeper investigation into:

- The rationale behind cloud-based policy enforcement
- Current NIST password security guidelines
- The evolution of password security best practices
- Implications for enterprise security planning

Key Takeaway

Even while focusing on SharePoint administration as the primary learning objective, related security observations provide valuable context for understanding the broader Microsoft 365 enterprise ecosystem. User identity and authentication mechanisms directly impact SharePoint permission models, making these foundational observations relevant to the core mission of this learning environment.

The following section provides detailed analysis of password policy differences and their security implications.

CRITICAL UNDERSTANDING: PERMISSION INHERITANCE AND MIGRATION

Permission Behavior During Content Migration

A fundamental principle of SharePoint administration that often creates confusion: **permissions configured in other Microsoft 365 environments do not transfer to SharePoint.** This applies regardless of how meticulously permissions are configured prior to migration.

Key Principle

When content is moved or copied from OneDrive (or any other source) into SharePoint:

- All existing permissions are discarded
- Content immediately inherits permissions from its destination location in SharePoint
- There is no "carry-over" of previous permission configurations

Why This Matters

This behavior requires administrators to understand that SharePoint permission management is a **distinct, intentional process** rather than an automatic inheritance from source systems. Even if an organization has:

- Established Azure AD security groups with specific access levels
- Configured explicit user permissions in OneDrive
- Set inheritance structures in the source environment

None of these configurations persist during migration to SharePoint.

Proper Workflow for SharePoint Permissions

1. **Plan SharePoint permission structure** based on organizational security requirements
2. **Migrate content** to appropriate SharePoint locations
3. **Break inheritance** where necessary (at site, library, folder, or file level)
4. **Apply permissions** according to SharePoint security design
 - Assign Azure AD groups to SharePoint permission levels
 - Configure explicit user permissions where required
 - Test access with different user roles

This "clean slate" approach ensures:

- Intentional security design rather than accidental permission inheritance
- Clear documentation of who has access to what
- Compliance with organizational security policies
- Elimination of permission conflicts from multiple sources

Lesson Learned

Do not invest time configuring complex permissions in OneDrive with the expectation they will transfer to SharePoint. Instead, design SharePoint permissions strategically from the beginning, understanding that migration creates an opportunity to implement proper access controls from a clean foundation.

PASSWORD POLICY COMPARISON: ON-PREMISES VS. CLOUD ENVIRONMENTS

Overview

Understanding password policy enforcement differences between on-premises Active Directory and cloud-based Azure AD is critical for enterprise security planning and implementation.

On-Premises Active Directory (Windows Server)

- **Administrative Control:** Full granular control over all password policies
- **Minimum Length:** Configurable down to zero characters (though not recommended)
- **Complexity Requirements:** Fully customizable or can be disabled
- **Use Case:** Allows simplified passwords for training environments (e.g., 2-character passwords)
- **Security Responsibility:** Organization assumes full liability for policy decisions

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Azure Active Directory (Microsoft 365)

- **Administrative Control:** Limited to strengthening baseline requirements
- **Minimum Length:** 8 characters (hard-coded, cannot be overridden)
- **Complexity Requirements:** Baseline requirements enforced by Microsoft
- **Rationale:** Microsoft maintains infrastructure security and compliance obligations
- **Security Responsibility:** Shared responsibility model—Microsoft enforces minimum standards

NIST SP 800-63B Current Guidelines

The National Institute of Standards and Technology provides the following modern password guidance:

Recommended Standards:

- **Minimum Length:** 8 characters (absolute minimum)
- **Best Practice Length:** 12-16+ characters for production environments
- **Composition:** Prioritize length over complexity
- **Password Expiration:** Eliminate mandatory periodic password changes unless compromise is detected
- **Multi-Factor Authentication (MFA):** Required for adequate security posture

Key NIST Findings:

- Long passphrases (e.g., "CorrectHorseBatteryStaple") provide superior security compared to short complex passwords (e.g., "P@ssw0rd1")
- Mandatory 90-180 day password rotation creates predictable patterns (Summer2024! → Fall2024! → Winter2025!)
- MFA implementation provides significantly more security value than password complexity alone

Outdated Practices (Pre-2017 NIST Guidelines):

- ~~Mandatory password changes every 90-180 days~~
- ~~Complex character requirements as primary security measure~~
- ~~Short passwords with special character substitution~~

Microsoft 365 E3 Security Posture Analysis

Microsoft's 8-character minimum password requirement meets NIST's absolute minimum standard but falls short of the recommended 12-16+ character best practice. This represents Microsoft's balance between:

- Security compliance requirements
- User experience and adoption
- Backward compatibility with legacy systems

Recommendation for Enterprise Deployment:

Organizations should implement the following enhanced security controls beyond Microsoft's defaults:

1. Enforce 16+ character passwords through conditional access policies where possible
2. Mandate MFA for all user accounts (Azure AD Premium feature)
3. Eliminate scheduled password expiration policies
4. Implement password breach detection and response protocols
5. Educate users on passphrase methodology over character substitution

Security Implication for SharePoint Administration

These password policies directly impact SharePoint security, as user authentication controls access to potentially sensitive documents and collaboration spaces. Understanding authentication mechanisms is foundational to implementing proper SharePoint permission structures.

CONDITIONAL ACCESS: ENTERPRISE TIME-BASED RESTRICTIONS

Note: This section represents current understanding based on initial exploration of Microsoft 365 E3 security capabilities. As practical implementation experience develops, this analysis may be refined or expanded.

Discovery Context

During exploration of Microsoft 365 E3 security capabilities, a question arose regarding the ability to restrict user access to corporate resources based on time of day—for example, limiting access to company data outside of business hours (9 AM - 5 PM). This is a common enterprise security requirement, particularly for organizations with:

- Remote workforce with company-issued devices
- Compliance requirements for work-hour restrictions
- Need to prevent after-hours data access for certain user groups
- Contractual obligations limiting access during specific time frames

Initial Research: Login Hours Feature

Traditional on-premises Active Directory includes a "Logon Hours" feature that allows administrators to restrict when users can authenticate to domain controllers. However, investigation revealed that this functionality is not available in cloud-only Azure Active Directory (Azure AD/Microsoft Entra ID) environments.

Critical Finding

Pure cloud Azure AD does not support native "logon hours" restrictions, regardless of license tier (Free, Premium P1, or Premium P2). This feature requires:

- On-premises Active Directory infrastructure
- Azure AD Connect with Pass-through Authentication (PTA)
- Hybrid identity configuration

For organizations operating entirely in the cloud (Microsoft 365 E3, no on-premises servers), the traditional "logon hours" approach is not applicable.

Enterprise Solution: Conditional Access Policies

Microsoft 365 E3 includes Azure AD Premium P1, which provides **Conditional Access**—a sophisticated enterprise security framework that enables time-based and location-based restrictions on corporate resource access.

Key Distinction

Conditional Access does not control:

- Local device login (Windows/macOS operating system authentication)
- Physical computer access

Conditional Access does control:

- Access to cloud applications (Outlook, Teams, SharePoint, OneDrive)
- Access to corporate data stored in Microsoft 365
- Authentication to enterprise resources

Practical Application

Scenario: Organization needs to restrict employee access to corporate email and documents outside of business hours.

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Implementation Strategy:

1. Navigate to **portal.azure.com** (Azure Portal)
2. Access **Microsoft Entra ID Security** → **Conditional Access**
3. Create policy with the following parameters: • **Users/Groups:** Select specific users or security groups requiring restrictions • **Cloud apps:** Select target applications (Exchange Online, SharePoint, Teams, etc.) • **Conditions:** Configure time-based or location-based access requirements • **Access controls:** Define action (Block access, Require MFA, etc.)

Real-World Example Use Cases

Time-Based Restriction:

- Block access to SharePoint after 6 PM and before 7 AM
- User attempting to access documents at 8 PM receives "Access Denied" message
- User can still use their device locally but cannot access corporate cloud data

Location-Based Restriction:

- Require Multi-Factor Authentication (MFA) for access from non-trusted IP addresses
- Block access entirely from specific countries or regions
- Allow unrestricted access only from corporate office networks

Combination Policies:

- Allow access during business hours from any location
- Require MFA for after-hours access from trusted locations
- Block all after-hours access from untrusted locations

Device Compliance Consideration

While basic Conditional Access policies do not require device enrollment in Microsoft Intune, organizations can create more sophisticated policies that check device compliance status:

- Is the device encrypted?
- Are security patches current?
- Is antivirus active and updated?

- Does the device meet minimum OS version requirements?

These device-based conditions require Microsoft Intune enrollment but provide additional security layers beyond time and location restrictions.

Lesson Learned

Cloud-based identity management operates fundamentally differently from traditional on-premises Active Directory. Rather than controlling device login authentication, enterprise security in Microsoft 365 focuses on controlling access to data and applications. This approach accommodates modern work patterns where employees may use personal devices, work remotely, or access resources from multiple locations—scenarios where traditional "logon hours" would be impractical or impossible to enforce.

Conditional Access represents a more sophisticated, flexible security model designed for cloud-first organizations. Understanding this distinction is critical for enterprise security planning and demonstrates the architectural differences between legacy and modern identity management systems.

Available for Testing

Conditional Access policies can be explored within the Microsoft 365 E3 trial environment, providing hands-on experience with enterprise security controls commonly deployed in production environments. The Azure Portal (portal.azure.com) provides access to these configuration options under Security → Conditional Access.

LAB CONTENT STRUCTURE DESIGN

Research and Planning

Prior to configuring SharePoint permissions and site architecture, a realistic content structure was developed to simulate an authentic penetration testing engagement environment. This approach ensures that permission testing reflects real-world security consulting scenarios rather than generic folder structures.

Methodology

Research was conducted on industry-standard penetration testing project organization, with particular attention to CREST methodology and professional engagement lifecycle management. The resulting structure mirrors the workflow and documentation requirements of actual security consulting engagements.

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Folder Structure Implementation

Eight primary categories were established, each representing a distinct aspect of a pen testing engagement:

Engagement Documents	Scope of Work
	NDA and Legal
	Rules of Engagement
	Project Time line
	Authorization Letters
	Contacts
Project Management	Meeting Notes
	Status Reports
	Scheduling
	Deliverables Tracking
Reconnaissance	OSINT
	Public Information
	Network Maps
	Asset Inventory
Enumeration	Service Enumeration
	Port Scans
	User Enumeration
	Technology Stack
Vulnerability Analysis	Findings
	Risk Ratings
	Evidence Placeholders
	False Positive Review
Reporting	Draft Report
	Final Report
	Executive Summary
	Remediation Recommendations
Internal Resources	Templates
	Style Guides
	Methodology
	Checklists
Archive	Old Engagements
	Version History
	Retired Documents

File Format Distribution

To simulate diverse document types commonly used in security engagements:

- Microsoft Word (.docx) - Reports, documentation, methodologies
- Microsoft Excel (.xlsx) - Tracking spreadsheets, findings logs, asset inventories
- Microsoft Visio (.vsdx) - Network diagrams, architecture maps

Initial Storage Location

All content was initially created and stored in OneDrive within the Microsoft 365 E3 trial environment. This staging approach allows for controlled migration into SharePoint site collections, enabling realistic testing of file migration scenarios and permission inheritance behaviors.

Rationale for Structure

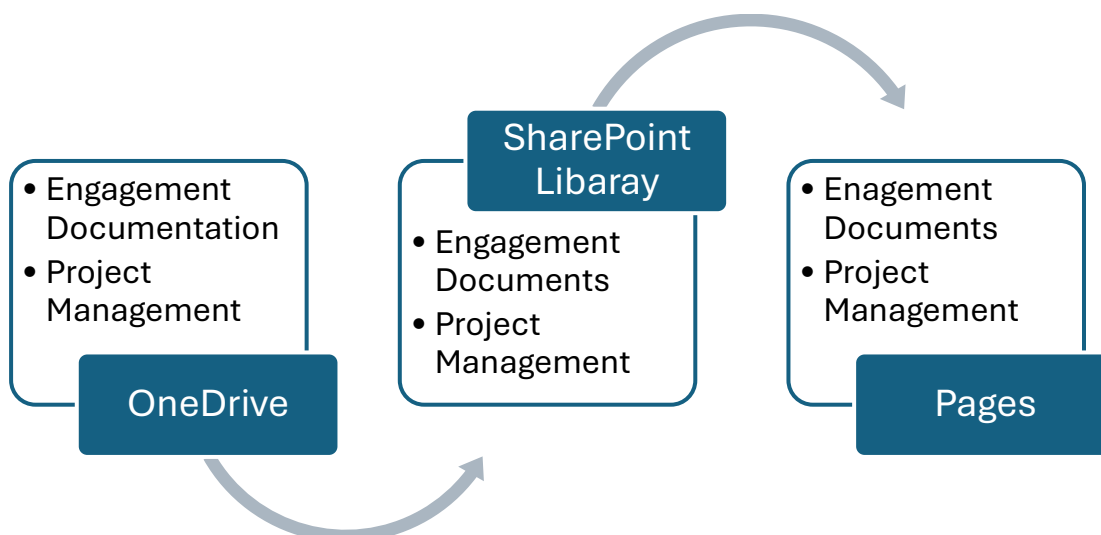
This organizational framework provides:

- **Realistic permission testing scenarios** - Different folders require different access levels (e.g., legal documents vs. technical findings)
- **Role-based access simulation** - Penetration testers, project managers, and clients would have varying permission requirements across these categories
- **Professional relevance** - Direct applicability to Security's operational structure

Next Steps

With content structure established, the following phases will be executed:

1. SharePoint site collection creation
2. Document library configuration
3. Migration of content from OneDrive to SharePoint
4. Permission model implementation and testing
5. User access validation across different roles



SHAREPOINT HOME PAGE NAMING CONSTRAINT

Critical Discovery

During site development, an attempt was made to rename the default Home page to match the site title ("Trula's Tech Trails"). This caused immediate navigation failure, resulting in 404 errors when accessing the site through standard navigation links.

Issue

SharePoint's default navigation structure expects the home page to be named "**Home.aspx**". While it is technically possible to rename page files, doing so for the Home page breaks the site's default navigation links, which are hardcoded to look for this specific filename.

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Resolution

The Home page filename must remain as "**Home**" in the Pages library. The site title/branding (displayed in headers, navigation bars, and hero sections) can be customized separately through page content and site settings, but the underlying page file must retain its original name.

Lesson Learned

Distinguish between:

- **Site title/branding** (customizable) - what users see
- **Page filenames** (constrained) - what SharePoint navigation references

Attempting to align these by renaming system pages will break core navigation functionality.

PERMISSION TESTING: ACCESS REQUEST WORKFLOW DISCOVERY

Initial Concern

During permission testing with user accounts assigned "Edit" permissions (Site Members group), it appeared that these users could freely share content and add new users to the site despite access request restrictions being enabled. This raised immediate security concerns about unauthorized access proliferation.

Testing Methodology

1. Configured site with "Allow members to share" disabled in Access Request Settings
2. Assigned TestLab2 user to "Site Members" group with Edit permissions
3. Logged in as TestLab2 and attempted to share content with TestLab6, TestLab7, and TestLab8 (users with no existing site access)
4. System allowed sharing actions to proceed without immediate blocking

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Critical Discovery

Further investigation revealed that the security mechanism was functioning correctly, but through an **approval workflow** rather than outright blocking:

What Actually Happens:

- Users with Edit & Read permissions can **initiate** sharing requests
- System displays message: "Sharing is limited on this item. You can only copy links for people who have existing access"
- Sharing request is **submitted** but not immediately granted
- **Site Owner receives email notification** requesting approval
- Shared users **cannot access content** until owner explicitly approves
- Owner can approve or decline each request individually

Validation Testing from Recipient Perspective:

To verify the security boundary, login testing was performed as TestLab8 (a user who had been "shared with" by both TestLab1 and TestLab2). Despite multiple sharing attempts by users with both Read and Edit permissions, TestLab8 was completely blocked from accessing the content. The system displayed "You do not have permission" with an option to request access. This confirms that sharing requests do not grant access - they only initiate an approval workflow. Content remains completely inaccessible to shared users until the Site Owner explicitly approves each request.

Why This Design is Effective

This workflow represents sophisticated enterprise access management:

Collaborative Efficiency:

- Team members can request access for legitimate collaborators (clients, partners, consultants)
- Workflow continues without waiting for IT/admin intervention for every single request

- Business operations maintain momentum

Security Control:

- Central authority (Site Owner) maintains final approval over all access
- Complete audit trail of who requested access and why
- Prevents unauthorized access - no one enters without explicit owner approval
- Owner can evaluate context and appropriateness of each request

Visibility and Accountability:

- All access requests documented
- Requestor identified in approval email
- Owner can communicate with requestor if clarification needed before approval

Real-World Application

In a penetration testing environment:

Scenario: Pentester needs client to review draft vulnerability report

Traditional Blocking Approach:

- Pentester must contact project manager/owner
- Owner manually adds client
- Delays in communication create project bottlenecks

Access Request Workflow:

- Pentester initiates share with client email
- Owner receives notification with context (who is requesting, for what)
- Owner approves (client gets immediate access to review draft)
- Maintains security while enabling efficient collaboration

Lesson Learned

Initial assumption that users could "freely share" was incorrect. The system implements **request-based sharing with mandatory approval**, which provides the optimal balance between:

- Operational efficiency (users can request access when needed)
- Security control (all access requires owner authorization)
- Audit compliance (complete record of all access requests and approvals)

This represents professional-grade access management more sophisticated than simple "allow/deny" permission models.

POWER AUTOMATE INTEGRATION: AUTOMATED APPROVAL WORKFLOWS

Discovery Context

During SharePoint administration exploration, the Power Automate integration capability was discovered through the "Automate" menu in document libraries. This integration enables automated business process workflows without custom code development, directly addressing organizational automation objectives.

Workflow Implementation

Flow Created: "Start approval when a new item is added"

Purpose: Automatically route newly uploaded documents through an approval workflow, notifying designated approvers and tracking approval status.

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Technical Configuration:

Trigger: "When an item is created"

- Monitors specified SharePoint document library
- Activates automatically upon file upload
- No manual intervention required

Action Sequence:

1. **Start and wait for an approval** - Sends approval request to designated approver(s)
 - Approval type: First to respond
 - Dynamic title includes filename
 - Includes file metadata (author, creation date)
 - Provides direct link to document
2. **Get my profile** - Retrieves approver information for email personalization
3. **Send an email** - Notifies requester of approval decision
 - Approval granted: Confirmation email sent
 - Approval rejected: Rejection notice with comments
4. **Terminate** - Completes workflow execution

User Experience:

For Document Uploader:

- Upload file to SharePoint as normal
- Workflow triggers automatically (no additional action required)
- Receives email notification of approval decision

For Approver:

- Receives email notification: "Please review: [filename]"
- Email contains:
 - Document metadata (who uploaded, when)
 - Direct link to file
 - **Approve/Reject buttons directly in email**

- Can approve/reject without leaving inbox
- Optional: Add comments explaining decision

Business Value for Security Consulting

Practical Application:

Scenario: Penetration test report workflow

1. Pentester completes vulnerability assessment draft
2. Uploads report to client project folder
3. Workflow automatically notifies project lead
4. Lead reviews report in context
5. Approves → Client receives notification report is ready
6. Rejects → Pentester receives feedback for revision

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Advantages over manual process:

- **Elimination of email bottlenecks** - No "Did you get my email?" follow-ups
- **Audit trail** - Complete record of all approval requests and decisions
- **Faster turnaround** - Approve directly from email notification
- **Consistency** - Every document follows same approval process
- **Visibility** - Status tracking shows where documents are in workflow

Strategic Alignment

This capability directly addresses organizational automation priorities communicated during pre-employment discussions. Power Automate integration with SharePoint enables:

- Business process automation without development resources
- Scalable workflow implementation across multiple engagement types
- Template-based approach (create once, apply to multiple scenarios)
- Integration with Microsoft 365 ecosystem (Teams, Outlook, Planner)

Lesson Learned

Power Automate transforms SharePoint from a static document repository into a dynamic business process platform. The ability to create approval workflows, notifications, and automated actions significantly enhances SharePoint's value for security consulting operations where document review cycles, client communications, and quality control processes are critical to service delivery.

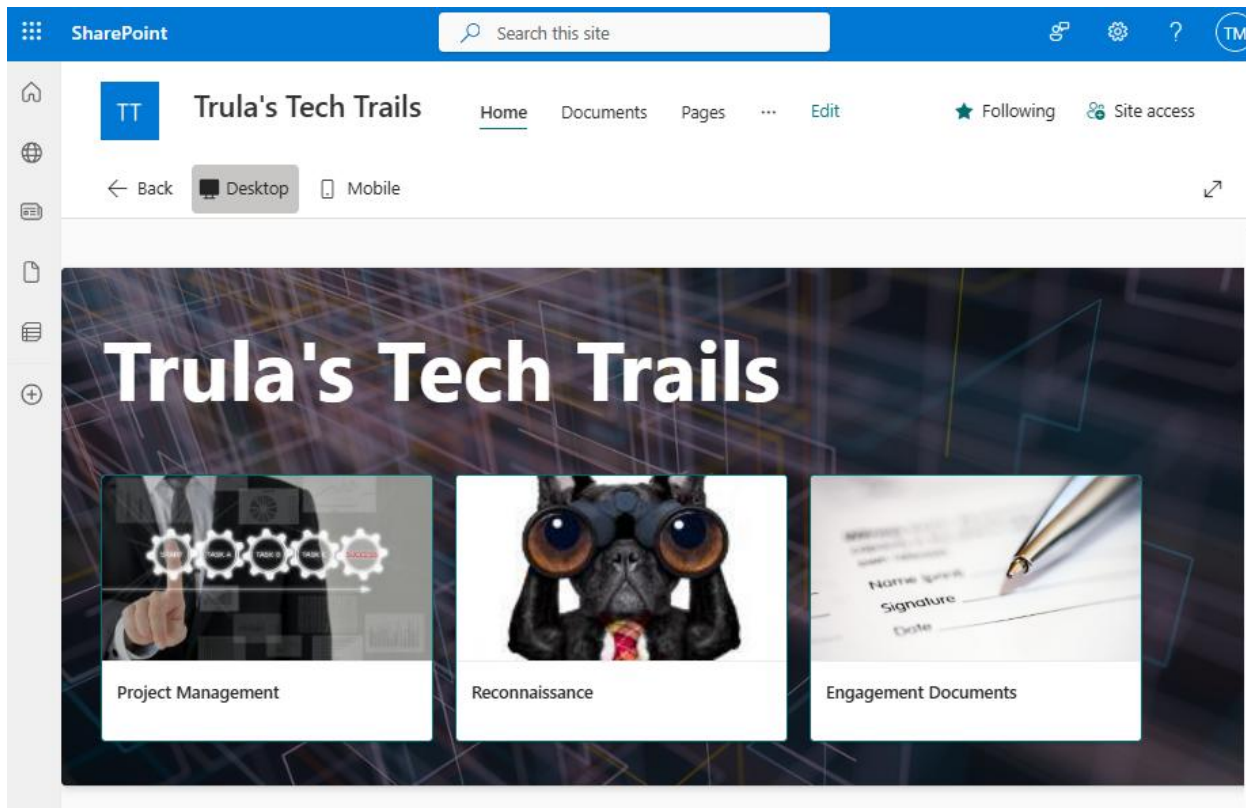
Future Exploration Opportunities

Additional Power Automate capabilities relevant to security consulting:

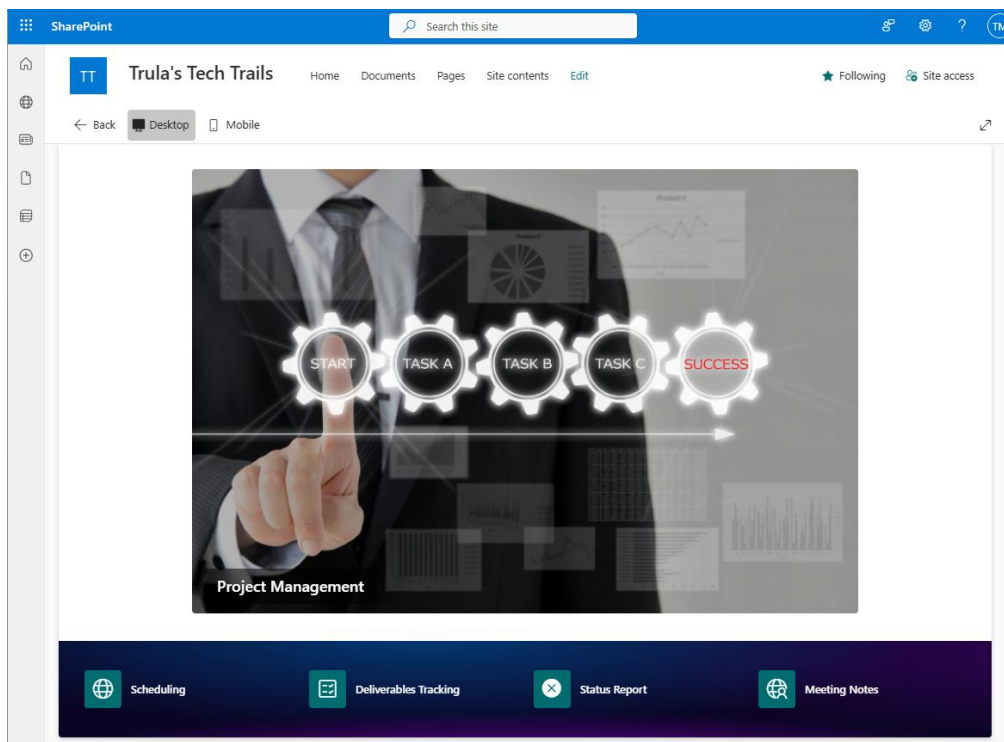
- Automated folder creation for new engagements
- Client notification when reports are published
- Scheduled reminders for pending approvals
- Integration with project management tools
- Custom notifications to Microsoft Teams channels

Screenshots

Trula's Tech Trails Home



Trula's Tech Trails Project Management page



Trula's Tech Trails Reconnaissance page

SharePoint Search this site

Trula's Tech Trails Home Documents Pages Site contents Edit

Following Site access

Back Desktop Mobile

Reconnaissance Intelligence Gathering Hub

Reconnaissance resources for information gathering and target analysis. Includes OSINT data, public information, network maps, and asset inventories essential for understanding the engagement environment.

Reconnaissance

- Asset Inventory
- Network Maps
- Public information
- OSINT

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Trula's Tech Trails Engagement page

SharePoint Search this site

Trula's Tech Trails Home Documents Pages Site contents Edit

Following Site access

Back Desktop Mobile

Engagement Documents

- Rules of Engagement
- Scope of Work
- Project Timeline
- Authorization Letter
- Contacts
- NDA and Legal

Trula's Tech Trails Document structure in SharePoint

SharePoint

Search this library

Trula's Tech Trails

Home Documents Pages Site contents Edit

Following Site access

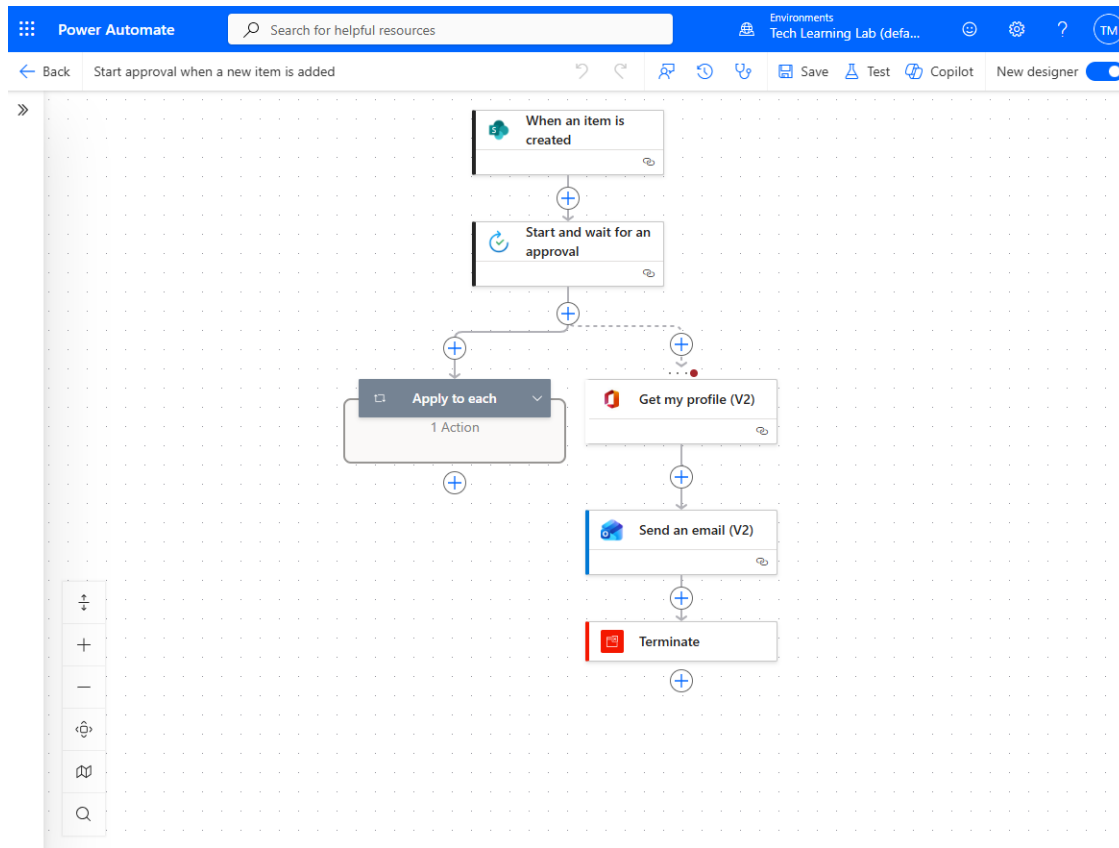
+ New Upload Edit in grid view Forms New Add shortcut to OneDrive Pin to Quick access ... All Documents Details

Documents

Name	Modified	Modified By	Sign-off status	Approval stat...	+ Add column
Archive	5 days ago	Trula McNease			
Engagement Documents	5 days ago	Trula McNease			
Enumeration	5 days ago	Trula McNease			
Internal Resources	5 days ago	Trula McNease			
Project Management	5 days ago	Trula McNease			
Reconnaissance	5 days ago	Trula McNease			
Reporting	5 days ago	Trula McNease			
Vulnerability Analysis	5 days ago	Trula McNease			

Trula's Tech Trails **POWER AUTOMATE INTEGRATION**

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Power Automate interface showing the flow details and run history.

Flows > Start approval when a new item is added

Details [Edit](#)

Flow	Status
Start approval when a new item is added	On
Primary owner	Created
Trula McNease	Jan 17, 2026, 05:23 PM
	Modified
	Jan 17, 2026, 05:24 PM
	Type
	Automated
	Plan
	This flow runs on owner's plan

[Original template](#)

28-day run history [Edit columns](#) [All runs](#)

Start	Duration	Status
Jan 17, 05:25 PM (16 h ago)	00:05:14	Succeeded

Savings ⓘ

We're sorry, but this feature is currently unavailable in your Dataverse organization. Please try again later. [Learn more](#)

Connection references [See all](#)

- SharePoint ✓
- Standard approvals ✓
- Office 365 Outlook ✓

[Show all](#)

Co-owner [Set primary owner](#) [Share](#)

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